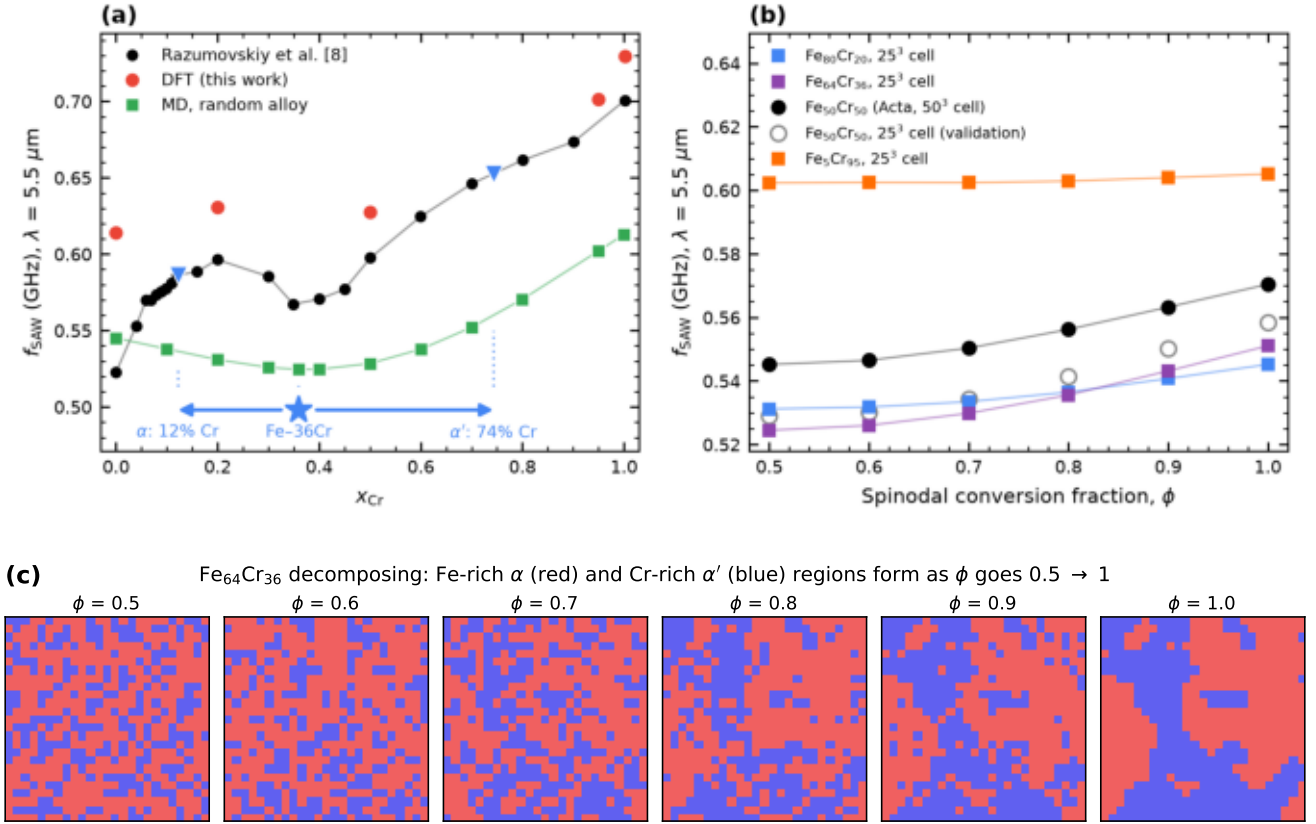

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
α: 12% Cr
Fe-36Cr
α′: 74% Cr
$f_{SAW}$ (GHz), λ = 5.5 μm
$x_{Cr}$
(b)
$Fe_{80}Cr_{20}$, $25^3$ cell
$Fe_{64}Cr_{36}$, $25^3$ cell
$Fe_{50}Cr_{50}$ (Acta, $50^3$ cell)
$Fe_{50}Cr_{50}$, $25^3$ cell (validation)
$Fe_5Cr_{95}$, $25^3$ cell
$f_{SAW}$ (GHz), λ = 5.5 μm
Spinodal conversion fraction, ϕ
(c)
$Fe_{64}Cr_{36}$ decomposing: Fe-rich α (red) and Cr-rich α′ (blue) regions form as ϕ goes 0.5 → 1
ϕ = 0.5
ϕ = 0.6
ϕ = 0.7
ϕ = 0.8
ϕ = 0.9
ϕ = 1.0
one atomic (001) plane of a $25^3$ bcc MD cell: red = Fe, blue = Cr